MIL-E-1/942D (See note 8)
13 April 1967
SUPERSEDING
MIL-E-1/942C
21 January 1959
(See note 9)

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 12BY7A

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pentode, miniature, sharp cutoff

Outline --- 6-3 (EIA)

Base --- E9-1

Envelope --- T6-1/2

Cathode --- Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8	9
Element	---	k	g1	g3 sd	h	h	hct	p	g2	g3 sd

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ec3	Ehk	Rk	Rg1	Ik	Pp	Pg2	TE	Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	v	Ohms	Meg	mAdc	W	W	°C	ft
Maximum:	6.9	330	0, -50	200	0	200	---	(see note 1)	45	7.2	1.2	200	(see note 2)
	13.9	---	---	---	---	---	---	---	---	---	---	---	---
Minimum:	5.7	---	---	---	---	---	---	---	---	---	---	---	---
	11.3	---	---	---	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	12.6	250	0	180	0	0	100	---	---	---	---	---	---

REQUIREMENT OR TEST:

GENERAL

3.2 Qualification - Required

E-50.2 Holding period

E-50.3 Preheating

5. Preparation for delivery

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METHOD OR PARAGRAPH	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1301	Heater current		0.65	II	If	275	325	ma
1266	Total grid current	See note 3	0.65	II	Ic1	0	-1.5	μAdc
1256	Electrode current (1) (plate)		0.65	II	Ib	31	31	mAdc
1256	Electrode current (2) (plate)	Ec1 = -16 Vdc	0.65	II	Ib	0	40	μAdc
1306	Transconductance (1)		0.65	II	Sm	8,500	13,500	μmhos
1201	Shorts and discontinuity detection		0.4	II	---	---	---	---
Appendix D, 20(a), 30	Visual and mechanical inspection criteria		---	---	---	---	---	---
Appendix D, 20(b)	Dimensions	Outline 6-3; dimensions A, C, D, and E	1.0	I	---	---	---	---
	Quality conformance inspection, part 2							
1211	Insulation of electrodes	E (g-all) = -100 Vdc E (p-all) = -300 Vdc	4.0	L6	R	10	---	Meg
1256	Electrode current (screen grid)		6.5	L6	Ic2	3.5	8.5	mAdc
1306	Transconductance (2)	Ef = 11.0 V	6.5	L6	Sm	8,000	---	μmhos
1336	Heater-cathode leakage		6.5	L6	Ink	---	20	μAdc
1331	Direct-interelectrode capacitance	No shield	6.5	L6	Cgp Cin Cout	--- 8.0 2.5	0.075 12.4 4.5	pf pf pf
1031	Low frequency vibration	Rp = 2,000 ohms; Ck = 1,000 μf	6.5	Code I	Ep	---	150	mVac
1041	Shock	G = 450; Ehk = 100 Vdc	---	---	---	---	---	---
1031	Vibration-fatigue test	G = 2.5; fixed frequency; F = 25 min, 60 max	6.5	See note 6	---	---	---	---
---	Post-shock and vibration-fatigue test end points	Low frequency vibration Heater-cathode leakage Total grid current Transconductance (1)	--- --- --- ---	--- --- --- ---	Ep Ihk Ic1 Sm	--- --- 0 7,000	200 50 -2.0 ---	mVac μAdc μAdc μmhos
2201	Noise and microphonics	Ef = 12.6 Vdc; Eg2 = Ep = 300 V; Rk = 100 ohms; Rg = 1 Meg; Rb = 0.004 Meg; Rg2 = 0.025 Meg; Ecal = 300 mV; C2g = 2 μf (min) (see note 7)	6.5	L6	---	---	---	---

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METHOD OR PARAGRAPH	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued							
1121	Base strain		---	---	---	---	---	---
2126	Glass strain		4.0	I	---	---	---	---
1106	Permanence of marking		---	---	---	---	---	---
---	Heater warmup time	See note 4	6.5	L6	t	6.8	17	sec

METHOD OR PARAGRAPH	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3					
4.7 and 1501	Life-test provisions (intermittent) (500 hours)	Group A; Ehk = 200 Vdc; Rg1 = 1.0 Meg; TE = 200° C (min) (see note 5)	---	---	---	---
4.7.3	Life-test end points (intermittent) (500 hours)	Total grid current	Ic1	0	-1.5	µAdc
		Heater-cathode leakage	Ihk	---	20.0	µAdc
		Insulation of electrodes	R	10	---	Meg
		Transconductance (1)	Sm	7,000	---	µmhos

NOTES:

1. Ec1 should never be positive; Rg1 = 1.0 megohm maximum for cathode bias or 0.25 megohm maximum for fixed bias.
2. See 6.5.2.3 for reduced pressure (altitude) rating, and for altitude, maximum peak voltage.
3. This test to be performed at the conclusion of the holding period.
4. Heater warmup time is defined as the time required for the voltage across the heater to reach 80 percent of the rated heater voltage after applying four times rated heater voltage to a circuit consisting of the tube heater in series with a resistance equal to three times the rated heater voltage divided by the rated heater current.
5. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (Method 1226), will be satisfied if a tube having bogey Ib (±5 percent) under normal test conditions, is determined to operate at minimum specified temperature at any position in the life-test rack.
6. This test shall be conducted on the initial lot and thereafter on a lot approximately every 6 months. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. When one lot has passed, the 6-month rule shall apply. MIL-STD-105, sample code letter E, shall apply.
7. The rejection level shall be set at the VU meter reading obtained during calibration.

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NOTES: -Continued

8. The tube described by this TSS is a substitute for MIL-E-1/942C, tube type 12BY7, and requires a new Federal Stock Number.

9. Changes from previous issue. The extent of changes (deletions, additions, etc.) precludes the annotation of the individual changes from the previous issue of this document.

Custodians:
Army - EL
Navy - SH
Air Force - 11

Review activities:
Army - EL
Navy - SH
Air Force - 11, 17, 85

User activities:
Army - MU
Navy - AS, OS, MC, CG
Air Force - 14, 19

Preparing activity:
Air Force - 11

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Code "C"

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